

Scalability & Future Plan

Date	8 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S. No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Lack of Prediction History Storage: HDI predictions and input indicators are processed in-memory and are not saved in a back-end database.	Researchers and policymakers cannot review past predictions, compare countries over time, or audit how the model classified specific inputs.	High
2	Stateless Single-Instance Deployment: The application runs on a single hosted instance with limited compute and memory capacity.	Under peak usage (e.g., multiple simultaneous users running predictions), requests can queue up, leading to higher latency or slower response times.	Medium
3	Static Offline Model Pipeline: The prediction model is trained offline and manually loaded into the application. It cannot adapt on its own.	The application cannot automatically adjust its classification logic as global development indicators (life expectancy, education, GNI per capita) evolve over time.	Medium

Scalability Plan:

S. No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Managed by a single hosted instance. It is prone to slower response times during concurrent request spikes.	Configure auto-scaling to automatically add instances when CPU or request load crosses a defined threshold, ensuring consistent response times under heavy usage.
2	Data Storage	No database persistence; predictions are computed and returned without being stored.	Integrate a managed database service (e.g., PostgreSQL or MongoDB Atlas) to persist country inputs, prediction results, and historical trends for later analysis.
3	Performance	Synchronous backend evaluation: every request runs the full prediction pipeline from scratch.	Implement a caching layer (e.g., Redis) to cache predictions for previously seen input combinations, reducing repeated computation.

4	Security	Open endpoint; performs basic input validation but lacks authentication or data encryption.	Add user authentication (e.g., JWT-based login) to protect prediction endpoints, and apply encryption for any stored sensitive data.
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Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Database Integration & Historical Dashboard	Month 3	Allows users to view historical HDI predictions and compare development trends across countries over time.
Phase 3	Continuous Model Retraining Pipeline	Month 6	Automatically retrains the model as updated global development datasets become available, keeping predictions accurate and current.
Phase 4	Explainable AI (XAI) Integration	Month 9	Integrates feature-contribution visualizations (e.g., SHAP values) to explain exactly which indicators most influenced a country's HDI classification.